CHURNGUARD

Customer Churn Prediction & Retention Strategy

Turning churn predictions into timely customer retention decisions

BUSINESS REPORT | XGBoost | Prediction App

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Audience: Business Leaders, Marketing and Operations Teams

Executive summary

What the findings mean for the business

Customers who churn

16.8%

About 17 out of every 100 customers in the dataset left.

Churners identified

94%

The model detects about 94 of every 100 actual churners.

Priority accounts

372

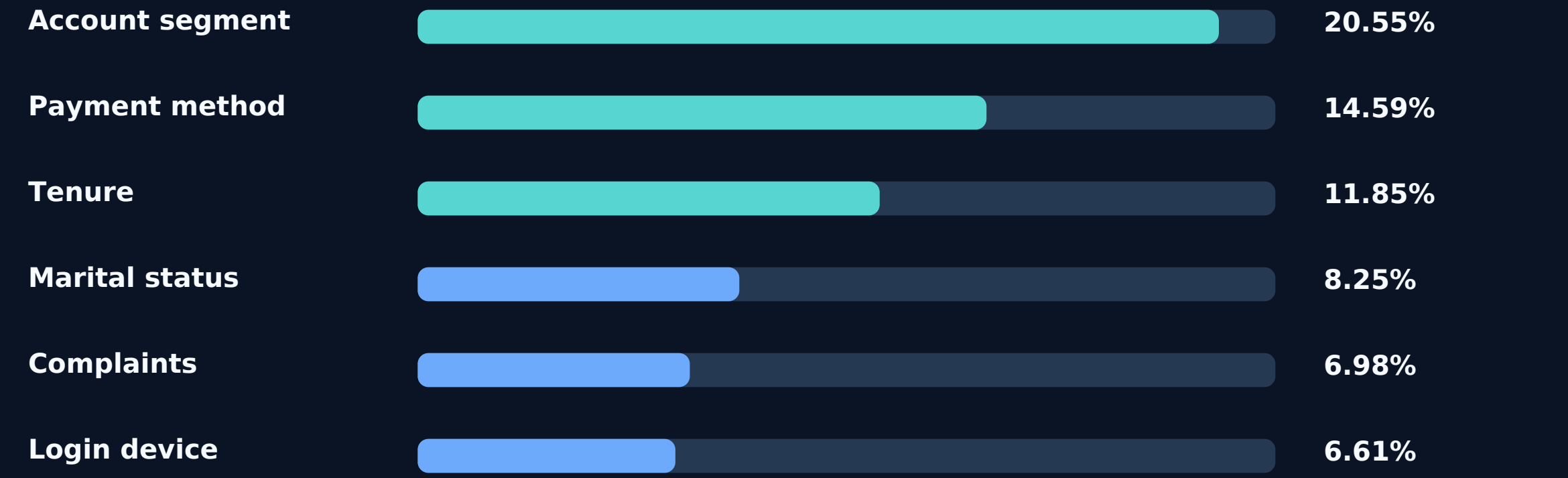
Critical and High-risk customers to target first.

Why this matters: Just 16.52% of accounts represent 78.60% of estimated monthly revenue exposure. Prioritizing these customers could improve retention efficiency.

Business decision: test targeted offers with a control group before committing to large-scale spending.

What drives the predictions?

Six leading factors used by the model — percentages indicate model importance



How to read this: Account segment contributes the most to model predictions (20.55%). These are measures of predictive importance — not the percentage of customers who churn, and not proof of cause.

How the early-warning system works

Email notifications are automatic when new predictions are made

1

Customer data

Unique Account ID + profile

2

Predict churn

Model estimates departure risk

3

Assign risk

Low / Moderate / High / Critical

4

Email alert

New High / Critical cases

5

Take action

Retention team follows up

Duplicate protection: The app records each notified Account ID and risk level in SQLite, so repeat predictions do not normally resend the same alert. Cloud restarts may reset local records.

Measured model performance

Test set: 2,252 customer accounts

Accuracy

96%

96 of 100 predictions are correct.

Recall

94%

Finds 94 of 100 actual churners.

Precision

86%

86 of 100 flagged customers churn.

F1-score

90%

Balances precision and recall.

PR-AUC

96%

Strong churner ranking despite imbalance.

ROC-AUC

99%

Strong separation of churn vs stay.

CV PR-AUC

94.81%

Consistent performance across validation folds.

Key interpretation: 94% recall means approximately 94 in every 100 customers who truly churn are identified.

Customer risk and revenue exposure

Monthly revenue at risk is probability-weighted; values are in dataset units

Critical	315 accounts	**68.36%**	**2,196.30**
High	57 accounts	**10.24%**	**329.08**
Moderate	118 accounts	**8.51%**	**273.54**
Low	1,762 accounts	**12.89%**	**414.00**

Critical + High: 372 accounts (16.52% of customers) hold 78.60% of estimated revenue exposure.

Business impact: where to focus

Concentrating outreach can direct limited resources to the greatest estimated risk

Priority customers

372

16.52% of the 2,252 evaluated accounts.

Share of exposure

78.60%

Nearly 79 of every 100 units of estimated revenue exposure.

Monthly exposure

2,525.38

Combined estimated exposure for Critical + High accounts.

Practical value: The retention team can act sooner through automatic emails and prioritize accounts with greater estimated revenue at risk. This is a decision-support benefit, not proven revenue saved.

Important: “Revenue exposure” is potential monthly revenue at risk, not revenue already lost.

Prediction App: what is available now

Operational tools that turn model scores into retention action

Single customer prediction

Enter an Account ID and customer details to estimate churn probability.

Batch customer prediction

Upload CSV/XLSX files and score multiple unique customer accounts.

Automatic Gmail alerts

Notify the retention team about newly identified High/Critical accounts.

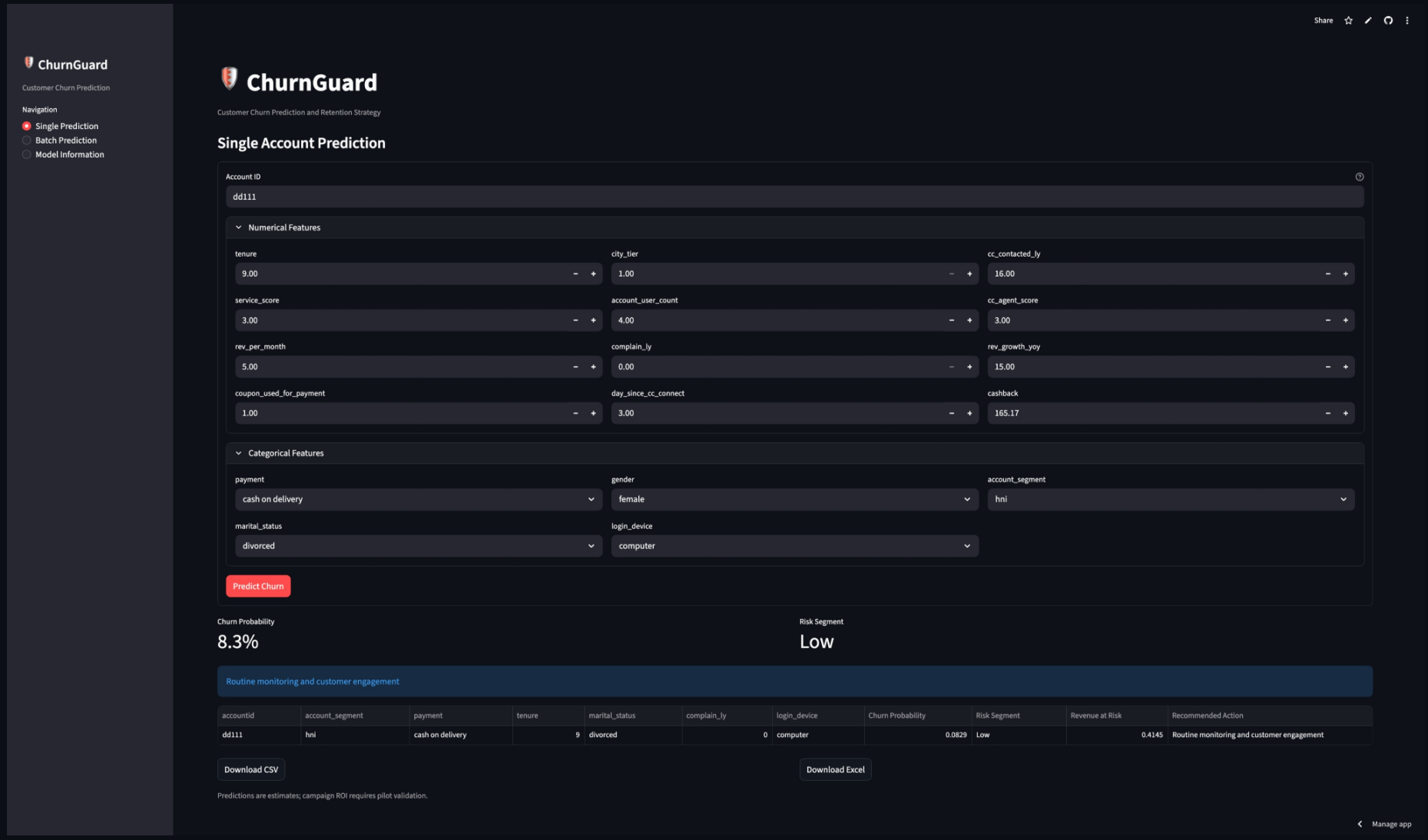
Results and exports

View risk groups, recommended actions and downloadable results.

Not yet included: scheduled background scoring or ongoing account intervention tracking.

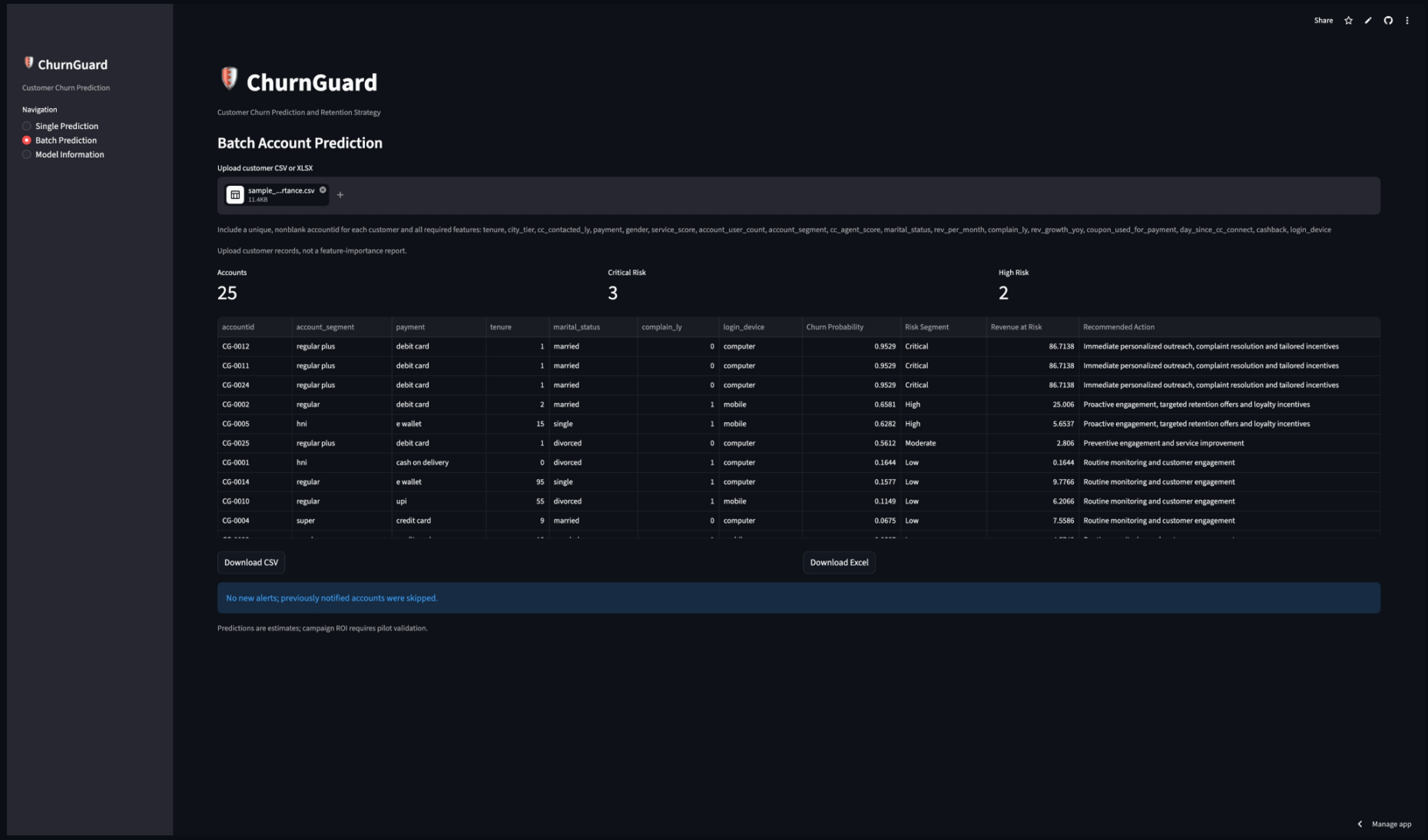
Prediction App — Single Account Prediction

Individual customer scoring, risk classification and recommended retention action



Prediction App — Batch Account Prediction

Bulk scoring, risk distribution, downloadable results and alert status



Retention scenarios: assumed success

Success rate = share of targeted at-risk customers retained because of an intervention

Conservative

Critical customers

15%

High-risk customers

20%

Base Case

Critical customers

30%

High-risk customers

40%

Optimistic

Critical customers

45%

High-risk customers

60%

Example: A 30% success assumption means 30 out of every 100 targeted Critical-risk customers are retained because of the offer. These are planning assumptions, not observed results.

Targeted retention offers

Recommended interventions, from lower to higher cost

01 **Personalized outreach**

Contact customers, resolve complaints and offer support.

5% cost

02 **Loyalty rewards**

Reward continued engagement when outreach alone is insufficient.

10% cost

03 **Discount vouchers**

Use targeted discounts for customers needing an added incentive.

15% cost

04 **Premium benefits**

Reserve for high-value accounts; conservative ROI may be negative.

60% cost

Offer costs are illustrative assumptions; compare results against control groups before scaling.

Projected NET benefit of retention

Base Case vs Optimistic | Personalized outreach at 5% cost

Base Case net benefit

2,705.53

Estimated retained contribution minus outreach cost.

Optimistic net benefit

4,128.47

Higher assumed retention success; same cost.

Combined outreach cost

140.35

Illustrative cost of targeting both priority segments.

What is net benefit? Expected retained revenue × 12 months × 30% contribution margin, less the offer cost. Figures are illustrative dataset units, not a confirmed currency. They are neither gross revenue nor realized profit.

Base assumptions: 30% Critical / 40% High success. Optimistic: 45% Critical / 60% High.

Business recommendations

Prioritize outreach, control incentives and test what works

01 **Focus on the 372 priority accounts**

They represent 78.60% of estimated revenue exposure.

02 **Start with affordable interventions**

Pilot outreach (5%), loyalty rewards (10%) and vouchers (15%).

03 **Act on automatic email alerts**

Contact newly flagged High/Critical accounts promptly.

04 **Maintain lighter-touch engagement**

Monitor 118 Moderate and 1,762 Low-risk customers.

05 **Reserve costly premium benefits**

Use 60% benefits only for exceptional, justified cases.

06 **Measure results before scaling**

Compare against control groups and calculate actual uplift/ROI.

Conclusion and next steps

Strong model results; business outcomes still need real-world validation

Delivered

XGBoost predictions, four risk segments, personalized recommendations, batch exports and automatic high-risk Gmail alerts.

Limitations

No verified campaign outcomes, intervention costs, churn timestamps or confirmed revenue units. Actual ROI remains unknown.

Future enhancement

Account tracking to follow changes in risk, retention actions and outcomes over time; also test persistent alerts and scale.

Recommended next step: run a controlled retention pilot, then expand only where benefits are measured.